

SHILATE

Software Hardware In Loop
Automotive Testing Environment

Tudor-Ciprian Tămaş

Supervisor: Sl. Dr. Ing. Valentin Mărănescu

Electronics, Telecommunications and Information Technology
Politehnica University of Timișoara

July 2026



The Industry Shift

- Vehicles are becoming **software-defined** — behaviour updated over-the-air
- Reinforcement learning (RL) is maturing as a practical control tool
- But no open-source pipeline bridges **RL training** and **SDV deployment**

Four Open Challenges

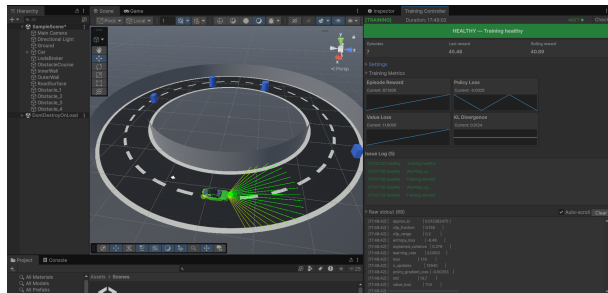
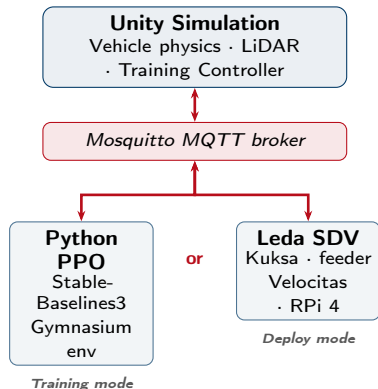
- 1 Simulation $\leftrightarrow$ middleware protocol coupling
- 2 Standardised signal representation (VSS)
- 3 Operational safety monitoring of learned policies
- 4 Reproducible, one-click training workflow

Goal

Build an **end-to-end, open-source** pipeline that:

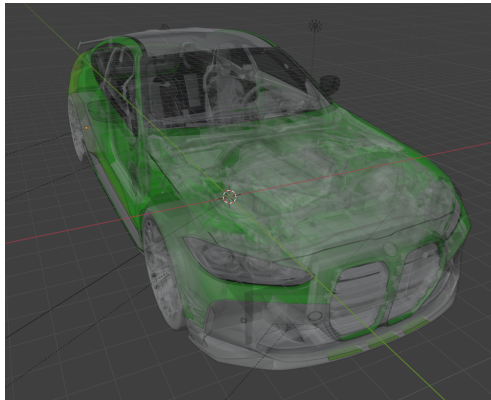
- Trains a deep RL agent in a Unity physics sim
- Deploys the policy onto a real Eclipse Leda SDV stack
- Uses the **same MQTT / VSS layer** from training to hardware

System Architecture — Training vs. Deploy Mode



Simulation environment overview

- **Training:** Unity Sim ↔ MQTT ↔ Python PPO
- **Deploy:** Unity Sim ↔ MQTT ↔ Eclipse Leda SDV
- Same **env0/** topic namespace in both modes — no protocol changes at deploy time



Rear-wheel-drive vehicle model

Vehicle & Sensor

- 1 500 kg RWD chassis — four WheelColliders, Pacejka tire model
- 21-ray LiDAR spanning 120° at 50 m range, published at 10 Hz
- Observation $o \in \mathbb{R}^{23}$: rays + normalised speed + steer

Reward Function

$$r_t = \underbrace{2 \Delta p_t}_{\text{progress}} + \underbrace{(-5) 1_{\text{collision}}}_{\text{collision}} + \underbrace{+50 1_{\text{lap}} + +25 1_{\frac{1}{2}\text{-turn}}}_{\text{milestones}}$$

Training Controller (Editor window)

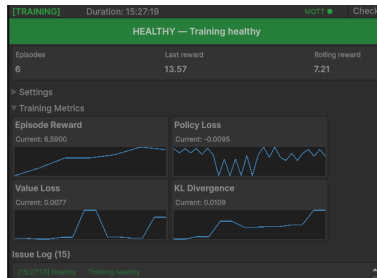
- One-click **Start Training / Run Model / Stop**
- Live health banner — 5 failure modes monitored
- Real-time reward, loss, and KL-divergence graphs

Algorithm & Architecture

- Proximal Policy Optimisation (PPO)
- Actor-Critic MLP: two hidden layers [256, 128]
- Action space: $a = [\delta, \theta, \beta]$ — steer $\in [-1, 1]$, throttle $\in [0, 1]$, brake $\in [0, 1]$

Key Hyperparameters

Parameter	Value
Total timesteps	100 000
Learning rate	3×10^{-4}
Rollout steps	2 048
Batch size	64
Clip ϵ / γ	0.2 / 0.999



Training convergence graphs (TensorBoard)

- Gymnasium-compliant wrapper — any SB3 algorithm can be substituted
- Structured telemetry: SHILATE-METRIC / SHILATE-HEALTH markers parsed live by the Editor
- 5-second observation timeout triggers health alert

```
[ OK ] Started Mosquitto MQTT Broker.
[ OK ] Started Eclipse Manto - Update Manager.
[ OK ] Started containers container runtime.
[ OK ] Started Eclipse Manto - Container Management.

Eclipse Leda v0.1.0-R3
leda-525400123456 login: root

[leda]

Eclipse Leda v0.1.0-R3 (GNU/Linux 5.15.124-yocto-standard x86_64)

Hostname:..... leda-525400123456
Uptime:..... 0 days, 0:0 hours
System load:..... 0.63 (1min) | 0.15 (5min) | 0.85 (15min)
Disk Space:..... Root: 698.7M | Used: 518.8M | Free: 128.1M
Data: 8.6G | Used: 4.8G | Free: 3.8G
RAM:..... Total: 2.8G | Used: 128.8M | Free: 3.6G
Network:..... not connected or activated
To check SDV services health, run $ sdv-health
Use $ sdu-device-info to display device configuration
Use $ sdv-provision to generate self-signed device certificates
```

Eclipse Leda container stack

5-Container Docker Compose Stack

- 1 **Mosquitto** — MQTT broker
- 2 **Kuksa databroker** — VSS signal store
- 3 **mqtt-kuksa-feeder** — MQTT → VSS bridge
- 4 **Velocitas app** — safety monitor
- 5 **leda-controller** — RL inference engine

Signal Chain

- env0/vehicle/speed → Vehicle.Speed (COVESA VSS)
- Velocitas app: overspeed alert (>120 km/h), RPM alert (>6 500)

Deployment to Raspberry Pi 4

- deploy-pi.sh — cross-compiles ARM64 via docker buildx, transfers and starts stack
- Full deploy cycle: ≈14 minutes

Remote Inference — Training Controller targets the Pi broker; no code changes, policy runs on-device

Training Convergence (3 seeds, mean)

Milestone	Timestep	σ
First collision-free episode	12 400	$\pm 2\,100$
Half-turn bonus triggered	31 000	$\pm 3\,800$
First full lap	48 500	$\pm 5\,200$
Rolling mean ≥ 60	76 000	$\pm 6\,900$

Inference — 50 episodes (local)

Metric	Value
Lap completion rate	97%
Mean episode reward	63.4
Mean lap duration	38.2 s
Mean speed (completed laps)	47.6 km/h
Collisions per episode	0.06

Ablation Study

- Half-turn bonus removed ($c_h = 0$): first lap delayed by $\approx 18\,000$ steps; rolling mean 12 pts lower at 100K
- Weaker collision penalty ($c_c = -1$): laps completed but agent drove closer to walls

Health Monitoring — All Failure Modes Detected

- Observation timeout (> 5 s) $\rightarrow$ *Critical*
- Python crash $\rightarrow$ non-zero exit code
- MQTT disconnect $\rightarrow$ *Disconnected* within 35 s

Raspberry Pi Remote Inference

Metric	Value
Lap completion rate (LAN)	95%
Additional latency (LAN)	15–20 ms
Deploy time	≈ 14 min

What was achieved

- End-to-end RL → SDV pipeline — same MQTT/VSS protocol from Unity training to Raspberry Pi 4 hardware
- Agent converges within 100 000 timesteps; 97% lap completion in inference
- One-click Training Controller with 5-failure-mode health monitoring
- Validated on physical hardware — 95% remote lap completion over LAN

Future Work

- Domain randomisation for sim-to-real transfer
- Safety monitor → active command override (not just alerts)
- Multi-agent scenarios and cooperative driving
- CAN bus hardware-in-loop testing on real ECUs



Thank you! Questions?